

# AYESHA ASLAM

*交通信息工程与控制*专业博士研究生。

*（预计 2027 年 1 月毕业）*

*机器学习与深度学习、智能交通、时间序列预测、Android 与 Web 开发、教学。*

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## 个人简介

我是长安大学（西安）交通信息工程与控制专业的博士研究生，预计 2027 年 1 月毕业。此前，我在 Bahria University（伊斯兰堡）获得计算机科学理学硕士学位，并在 University of the Punjab（拉合尔）获得计算机科学理学学士学位，由此打下了扎实的计算机科学基础。我的研究聚焦四个方向：自动驾驶车辆、轨迹预测、路径规划和无人系统，研究成果已以期刊论文、会议论文和书籍章节的形式，由 Springer Nature、Elsevier、IEEE 和 CRC Press 出版。我还在伊斯兰堡的学校任教计算机科学三年以上，并持有教育学学士学位。工作之余，我喜欢素描和油画。

## 教育经历

- 博士，*交通信息工程与控制* — **2022/09 –** 预计 **2027/01** 毕业
  *长安大学，中国西安*
- 理学硕士，*计算机科学（**MSCS**）* — **2019/09 – 2022/01**
  *Bahria University, Islamabad, Pakistan*
- 教育学学士（**B.Ed**） — **2017/09 – 2019/01**
  *Allama Iqbal Open University, Islamabad, Pakistan*
- 理学学士，*计算机科学（**BSCS**）* — **2011/09 – 2016/01**
  *University of the Punjab, Lahore, Pakistan*

## 工作经历

- 计算机科学教师 — **2020/10 – 2022/11**
  *Army Public School, Pothwar Campus, Islamabad, Pakistan*
- 教师 — **2019/03 – 2020/03**
  *Dar-e-Arqam School, G-9/2, Islamabad, Pakistan*
- 教师 — **2018/09 – 2019/02**
  *SLS School, G-11/4 Campus, Islamabad, Pakistan*
- 人力资源经理兼开发实习生 — **2017/09 – 2018/02**
  *Itroos, Islamabad, Pakistan*
- **Android** 开发工程师（实习） — **2017/03 – 2017/08**
  *Xyper Technologies, Islamabad, Pakistan*

## 精选论文

详见附录或 Google Scholar 查看完整论文列表。
* 通讯作者；** 共同通讯作者。

1. **Ayesha Aslam**, Xiaojun Yang. Spatial-aware self-attention feature fusion DQN for path planning of USV in static and dynamic environments. Applied Soft Computing (2026), 115818, Elsevier.
2. **Ayesha Aslam**, Yang Xiaojun*, Kashif Naseer Qureshi**, Adil Hussain, Kayhan Zrar Ghafoor. GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction. Journal of Engineering Research (2025).
3. Tariq, Zhu Wei, Kamran Shaukat*, **Ayesha Aslam**, Adil Hussain, Md. Mahboob Alam**. A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning. Scientific Reports (2026).
4. Adil Hussain*, Vishwanath Eswarakrishnan**, **Ayesha Aslam**, Sajib Tripura. Charging stations demand forecasting using LSTM based hybrid transformer model. Scientific Reports (2025).
5. **Ayesha Aslam**, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, Adil Hussain. Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh (2025).
6. **Ayesha Aslam**, Adil Hussain, Noman Iqbal, Muhammad Yasir, Sajib Tripura. Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen (2025).
7. **Ayesha Aslam**, Sajib Tripura, Muhammad Rabiu Abdullahi, Vineet Dhanawat, Adil Hussain. Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer (ViT) Model. 3rd International Conference on Data Science and Emerging Technologies (DaSet), Springer, Malaysia (2024).
8. **Ayesha Aslam**, Adil Hussain, Kashif Naseer Qureshi. Big Data Analytics for AI in Renewable Energy Networks (Chapter 9). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier (2025).

## 技能与专长

- 教学：计算机科学教学、课程与教案设计、课堂管理、学生评估与反馈、学生指导（教育学学士；三年以上学校教学经验）。
- 教学兴趣：计算机科学、程序设计、人工智能与机器学习、智能交通系统。
- 编程语言：Java, C#, C++, C
- 开发工具：Android Studio, Visual Studio, NetBeans, MS SQL, MySQL
- 操作系统：Windows, Ubuntu, Kali Linux
- 科研工具：EndNote, TeXMaker, MiKTeX, JabRef, Gnuplot, Inkscape
- 办公软件：Word, PowerPoint, Excel, OneNote
- 优势：教学、科研、Android 开发、人力资源管理
- 兴趣爱好：素描与油画

## 精选项目

- 学校管理系统（**Dar-e-Arqam School**，费萨拉巴德）：为多校区打造一体化平台，涵盖招生、成绩、学费单与收缴、工资、考勤和日记账。
- **ADMS**，律师案件日志管理系统（**Pansoota Law Associates**）：交付定制的 MVC 案件日志系统，包含庭审日程、法院与客户管理、付款跟踪和短信提醒，并提供基于 CMS 的公开网站。
- **SmrtWorx**：设计该应用并交付其功能需求，包括多国货币转换器和汇款订单处理器。
- 求职门户（**Job Portal**）：开发面向求职者、雇主和管理员的门户，支持免费与高级账户、直接招聘和个人资料推广。
- **Assignment Helps**：开发学生辅导应用，连接学习者与澳大利亚专家。
- **I Secure**：开发 Java Android 应用，作为毕业设计。

*在线项目链接可应要求提供。*

## 推荐人

可应要求提供。

## 附录：完整论文列表

*所有期刊论文、会议论文和书籍章节列表*

* 通讯作者；** 共同通讯作者。

### 期刊论文

**2026**

1. **Ayesha Aslam**, Xiaojun Yang. Spatial-aware self-attention feature fusion DQN for path planning of USV in static and dynamic environments. Applied Soft Computing (2026), Article 115818, Elsevier.
2. Tariq, Zhu Wei, Kamran Shaukat*, **Ayesha Aslam**, Adil Hussain, Md. Mahboob Alam**. A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning. Scientific Reports.

**2025**

3. Adil Hussain*, Vishwanath Eswarakrishnan, **Ayesha Aslam**, Sajib Tripura. Charging stations demand forecasting using LSTM based hybrid transformer model. Scientific Reports (2025).
4. **Ayesha Aslam**, Yang Xiaojun*, Kashif Naseer Qureshi**, Adil Hussain, Kayhan Zrar Ghafoor. GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction. Journal of Engineering Research.
5. **Ayesha Aslam**, Z. D. Anuarovna. Real-Time Workload Prediction and Resource Optimization for Parallel Heterogeneous High-Performance Computing Systems Architectures. Ubiquitous Technology Journal, 1(1), 40-47.

**2024**

6. Adil Hussain*, **Ayesha Aslam**, Sajib Tripura, Vineet Dhanawat, Varun Shinde. Weather Forecasting Using Machine Learning Techniques: Rainfall and Temperature Analysis. Journal of Advances in Information Technology.
7. Adil Hussain*, **Ayesha Aslam**. Hate Speech against Women and Immigrants: A Comparative Analysis of Machine Learning and Text Embedding Techniques. Journal of Applied Research and Technology.
8. Adil Hussain*, **Ayesha Aslam**. Ensemble-based Approach using Inception V2, VGG-16, and Xception Convolutional Neural Networks for Surface Cracks Detection. Journal of Applied Research and Technology.
9. Adil Hussain, Amna Khatoon*, **Ayesha Aslam**, Tariq, Muhammad Asif Khosa. A Comparative Performance Analysis of Machine Learning Models for Intrusion Detection Classification. Journal of Cyber Security.
10. **Ayesha Aslam**, Adil Hussain*. A Performance Analysis of Machine Learning Techniques for Credit Card Fraud Detection. Journal of Artificial Intelligence.
11. Adil Hussain*, **Ayesha Aslam**. Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models. Journal of Artificial Intelligence.

**2023**

12. Adil Hussain, Asad Ullah*, **Ayesha Aslam**, Amna Khatoon. A Modified Siamese Network for Facial Assimilation. WSEAS Transactions on Signal Processing.

## 会议论文

**2026**

1. Adil Hussain*, **Ayesha Aslam**. A Reinforcement Learning-Based Adaptive Dynamic Pricing for EV Charging Stations. World Transport Congress (WTC), Tianjin, China.
2. Sajib Tripura, Noman Iqbal, Zahirul Islam Kajal, Adil Hussain, **Ayesha Aslam**, Mostak Ahamed. Traffic Speed Prediction using LSTM and Spatial GCN-based Feature Fusion. IEEE 3rd International Conference on Computing, Applications and Systems, Bangladesh (2026) - ACCEPTED.

**2025**

3. Sajib Tripura, Adil Hussain, Mostak Ahamed, Dhonita Tripura, **Ayesha Aslam**. A Dual Attention Mechanism-Based Enhanced Residual Network for Skin Cancer Detection. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh.
4. **Ayesha Aslam**, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, Adil Hussain. Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting. IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS), Bangladesh.
5. Adil Hussain, Muhammad Yasir, Noman Iqbal, Sajib Tripura, **Ayesha Aslam**. Heart Disease Classification Using Machine Learning Techniques: An Impact Analysis of GridSearchCV-based Optimization. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
6. **Ayesha Aslam**, Adil Hussain, Noman Iqbal, Muhammad Yasir, Sajib Tripura. Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.

**2024**

7. Adil Hussain*, **Ayesha Aslam**. A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot. World Transport Congress (WTC), Qingdao, China.
8. Adil Hussain, Vineet Dhanawat, **Ayesha Aslam**, Noman Iqbal, Sajib Tripura. Credit Card Fraud Detection Using Machine Learning Techniques: Dealing with Imbalanced Data Using Over-Sampling and Under-Sampling Methods. IEEE Beyond Technology Summit on Informatics International Conference (BTS-I2C), Indonesia.
9. Adil Hussain, Kashif Naseer Qureshi, Raja Waseem Anwar, **Ayesha Aslam**. A Novel SCD11 CNN Model Performance Evaluation with Inception V3, VGG16 and ResNet50 Using Surface Crack Dataset. IEEE 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS), Oman.
10. Adil Hussain, Kashif Naseer Qureshi, Faizan Zaman, **Ayesha Aslam**. Minor Surface Cracks Detection using SCD11 Convolutional Neural Network. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
11. Adil Hussain, Kashif Naseer Qureshi, **Ayesha Aslam**, Tariq, Muhammad Rabiu Abdullahi. TSC18 Convolutional Neural Network for Traffic Sign Classification. IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA), Yemen.
12. Adil Hussain, Vineet Dhanawat, **Ayesha Aslam**, Faizan Zaman. Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model. IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET), Malaysia.
13. **Ayesha Aslam**, Sajib Tripura, Muhammad Rabiu Abdullahi, Vineet Dhanawat, Adil Hussain. Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer (ViT) Model. 3rd International Conference on Data Science and Emerging Technologies (DaSet), Springer, Malaysia.

14. Faizan Zaman, Zhigang Xu, Adil Hussain, **Ayesha Aslam**, Muhammad Rabiu Abdullahi. Concrete and Pavement Cracks Detection Leveraging Vision Transformer (ViT) Model. IEEE International Conference on Electrical and Information Technology (IEIT), Indonesia.

**2023**

15. Adil Hussain, **Ayesha Aslam**, Tariq. Traffic Sign Classification Using a Novel TSC18 Convolutional Neural Network. 9th International Conference of Sabaragamuwa University of Sri Lanka (Abstract), Sri Lanka.

## 书籍章节

**2026**

1. Adil Hussain, **Ayesha Aslam**, Mahad Ahmad. Artificial Intelligence in Agriculture: Applications, Challenges, and Future Directions (Chapter 3). AI-Driven Farming Systems With Minimal Human Intervention, 1st Edition, IGI Global, eISBN13: 9798337392974.

**2025**

2. Adil Hussain, **Ayesha Aslam**, Rundong Wang, Kashif Naseer Qureshi. Advanced Learning Techniques for AI in Renewable Energy (Chapter 7). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier, eBook ISBN: 9780443406195.
3. **Ayesha Aslam**, Adil Hussain, Kashif Naseer Qureshi. Big Data Analytics for AI in Renewable Energy Networks (Chapter 9). Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks, 1st Edition, Elsevier, eBook ISBN: 9780443406195.
4. Adil Hussain, Kashif Naseer Qureshi, **Ayesha Aslam**. Novel Machine and Deep Learning and Training Techniques for AI-based Healthcare Systems (Chapter 7). Artificial Intelligence-based Smart Healthcare Systems: New Standards, Technologies and Communication Systems, 1st Edition, Elsevier, Paperback ISBN: 9780443264764.
5. **Ayesha Aslam**, Kashif Naseer Qureshi, Adil Hussain. Advanced AI-Based Healthcare Systems Applications and Services (Chapter 4). Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns, 1st Edition, Taylor & Francis - CRC Press, eBook ISBN: 9781032667911.